Exoplanet Hunters: Analysis of STAC-1, STAC-2 and STAC-3

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***Abstract*—An exoplanet is a planet that orbits a star other than the Sun. When a planet passes in front of its host star, the star's observed brightness decreases slightly. This is called a transit. We analysed the relative brightness of three stars, STAC-1, STAC-2 and STAC-3, and found a transiting exoplanet around each one. STAC-1 b is Neptune-sized and orbits very close to its star. STAC-2 b is also Neptune-sized. STAC-3 b is a Super-Earth. By comparing each orbital distance with the habitable zone of its star, we found that STAC-3 b is the most promising candidate for further study.**

***Index Terms*—exoplanet, transit, orbital period, habitable zone**

I. Assumptions, Formulas and Stellar Data

A. Assumptions

1) The planet moves in a circular orbit around its host star.
2) The orbit is seen edge-on, so the planet crosses the middle of the star.
3) The planet is dark and blocks all the light behind it.
4) The star is equally bright across its whole surface.
5) The mass of the planet is very small compared with the mass of the star.
6) The orbital period is taken from the time between repeated transits.
7) The habitable-zone relation ignores clouds and the atmosphere of the planet.

B. Formulas

$$\delta = \frac{F_{\text{normal}} - F_{\text{transit}}}{F_{\text{normal}}} \tag{1}$$

$$\delta \approx \left(\frac{R_p}{R_s}\right)^2 \;\Rightarrow\; R_p = R_s\sqrt{\delta} \tag{2}$$

$$P^2 = \frac{a^3}{M} \;\Rightarrow\; a = \left(P^2 M\right)^{1/3} \tag{3}$$

$$d_{\text{HZ}} \approx \sqrt{L/L_\odot}\ \text{AU} \tag{4}$$

Meaning of the symbols:

- δ = transit depth.
- F_{normal} = brightness outside transit; F_{transit} = brightness during transit.
- R_p = radius of the planet; R_s = radius of the star.
- P = orbital period in years; a = orbital distance in AU.
- M = mass of the star in solar masses.
- d_{HZ} = approximate distance of the habitable zone from the star.
- L = luminosity of the star; $L_\odot$ = luminosity of the Sun.

Constants: $R_\odot = 696{,}000$ km, $R_\oplus = 6371$ km, 1 AU $= 149.6$ million km, 1 year $= 365.25$ days.

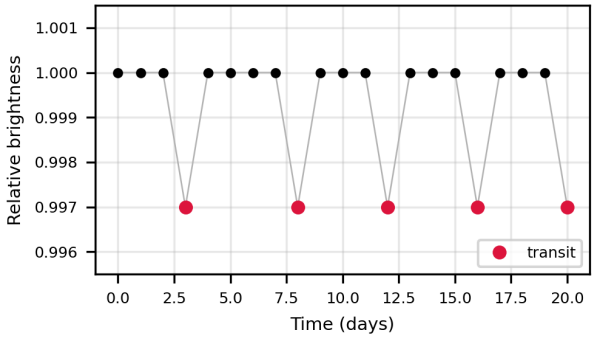


Fig. 1. Relative brightness of STAC-1. Red points are transits.

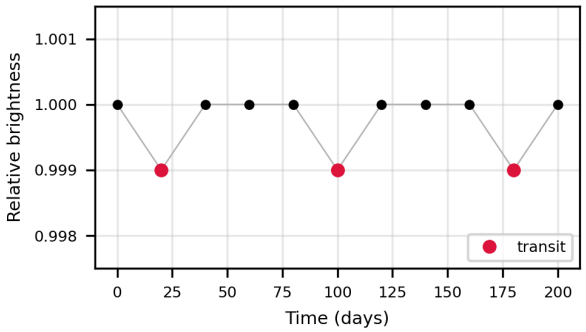


Fig. 2. Relative brightness of STAC-2. Red points are transits.

C. Stellar Properties

TABLE I
Stellar properties (given)

Property	STAC-1	STAC-2	STAC-3
Mass	1.00 $M_\odot$	0.80 $M_\odot$	0.50 $M_\odot$
Radius	1.00 $R_\odot$	0.80 $R_\odot$	0.50 $R_\odot$
Luminosity	1.00 $L_\odot$	0.40 $L_\odot$	0.08 $L_\odot$
Surface temperature	5800 K	5000 K	3800 K

II. Part A: Detect the Exoplanets

A. Q1: Do the Observations Indicate a Transiting Exoplanet?

The relative brightness outside transit is 1.0000. Any value lower than this is a decrease in brightness. These are marked in Figs. 1–3 and listed below.

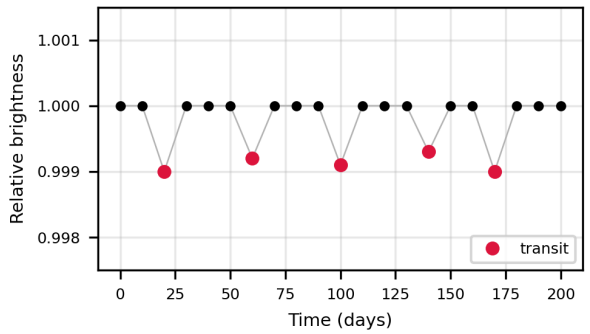


Fig. 3. Relative brightness of STAC-3. Red points are transits.

TABLE II
TIMES WHEN THE RELATIVE BRIGHTNESS DECREASED

Star	Time (days)	Periodic?
STAC-1	3, 8, 12, 16, 20	Yes
STAC-2	20, 100, 180	Yes
STAC-3	20, 60, 100, 140, 170	Yes

Answer: Yes. All three stars show a repeated decrease in brightness of about the same size, so each star has a transiting exoplanet.

B. Q2: Approximate Orbital Period

The orbital period is the time between one transit and the next.

STAC-1

Working:

$$8-3=5, \quad 12-8=4,$$
$$16-12=4, \quad 20-16=4 \text{ days}$$

Three intervals are 4 days. The first interval is 5 days, a little different. The average of all four intervals is $(20-3)/4 = 4.25$ days.

Answer: $P_1 \approx 4$ days.

STAC-2

Working:

$$100-20=80, \quad 180-100=80 \text{ days}$$

Observations are only every 20 days, so a transit may happen between two observations. At $t = 40, 60, 80, 120, 140, 160$ days there is no transit, so the orbital period cannot be 20 or 40 days. A shorter period cannot be completely ruled out, but 80 days is the simplest value that fits all the data.

Answer: $P_2 \approx 80$ days.

STAC-3

Working:

$$60-20=40, \quad 100-60=40,$$
$$140-100=40, \quad 170-140=30 \text{ days}$$

Three intervals are 40 days. The last is 30 days. The average of all four is $(40+40+40+30)/4 = 37.5$ days.

Answer: $P_3 \approx 40$ days.

C. Q3: Transit Depth

Formula: $\delta = (F_{\text{normal}} - F_{\text{transit}})/F_{\text{normal}}$, with $F_{\text{normal}} = 1.0000$.

Working:

$$\delta_1 = \frac{1.0000 - 0.9970}{1.0000} = 0.0030$$
$$\delta_2 = \frac{1.0000 - 0.9990}{1.0000} = 0.0010$$
$$\delta_3 = \frac{0.0010+0.0008+0.0009+0.0007+0.0010}{5} = 0.00088$$

The five transits of STAC-3 have slightly different depths, so we used the average.

Answer: $\delta_1 = 0.30\%$, $\delta_2 = 0.10\%$, $\delta_3 = 0.088\%$.

D. Q4: Why a Periodic Decrease in Brightness Indicates an Exoplanet

When a planet passes in front of its host star, it blocks a small part of the star's light, so the observed brightness of the star decreases slightly. After the transit the brightness returns to normal. The planet keeps orbiting, so the transit happens again after every orbit. A decrease that repeats at regular times and always has the same depth is unlikely to be random, so it is a strong sign of a planet in orbit. The time between transits gives the orbital period, and the transit depth gives the size of the planet.

TABLE III
PART A RESULTS

	STAC-1	STAC-2	STAC-3
Exoplanet detected?	Yes	Yes	Yes
Orbital period (days)	4	80	40
Transit depth δ	0.0030	0.0010	0.00088

III. PART B: DETERMINE THE PLANET SIZE

A. Q5: Radius of Each Planet in Kilometres

Formula: $R_p = R_s\sqrt{\delta}$.

STAC-1 b

Working:

$$R_s = 1.00 \times 696{,}000 = 696{,}000 \text{ km}$$
$$\sqrt{\delta} = \sqrt{0.0030} = 0.0548$$
$$R_p = 696{,}000 \times 0.0548 = 38{,}100 \text{ km}$$

STAC-2 b

Working:

$$R_s = 0.80 \times 696{,}000 = 556{,}800 \text{ km}$$
$$\sqrt{\delta} = \sqrt{0.0010} = 0.0316$$
$$R_p = 556{,}800 \times 0.0316 = 17{,}600 \text{ km}$$

STAC-3 b

Working:

$$R_s = 0.50 \times 696{,}000 = 348{,}000 \text{ km}$$
$$\sqrt{\delta} = \sqrt{0.00088} = 0.0297$$
$$R_p = 348{,}000 \times 0.0297 = 10{,}300 \text{ km}$$

Answer: STAC-1 b $\approx$ 38,100 km, STAC-2 b $\approx$ 17,600 km, STAC-3 b $\approx$ 10,300 km.

B. Q6: Radius in Terms of Earth's Radius

Formula: $R_p/R_\oplus$, with $R_\oplus = 6371$ km.

Working:

$$\text{STAC-1 b: } 38{,}100/6371 = 5.98$$
$$\text{STAC-2 b: } 17{,}600/6371 = 2.76$$
$$\text{STAC-3 b: } 10{,}300/6371 = 1.62$$

Answer: STAC-1 b $\approx 5.98\,R_\oplus$, STAC-2 b $\approx 2.76\,R_\oplus$, STAC-3 b $\approx 1.62\,R_\oplus$.

C. Q7: Classification of Each Planet

The problem does not give exact size limits, so we used these approximate ones:

TABLE IV
SIZE CLASSES USED

Class	Radius
Earth-sized	up to 1.25 $R_\oplus$
Super-Earth	1.25 to 2 $R_\oplus$
Neptune-sized	2 to 6 $R_\oplus$
Jupiter-sized	more than 6 $R_\oplus$

STAC-2 b ($2.76\,R_\oplus$) is above the Super-Earth limit, so it is Neptune-sized. STAC-1 b ($5.98\,R_\oplus$) is just under the Jupiter-sized limit, so it is a large Neptune-sized planet.

Answer: STAC-1 b: Neptune-sized (large). STAC-2 b: Neptune-sized (small). STAC-3 b: Super-Earth.

IV. PART C: DETERMINE THE PLANETARY ORBITS

A. Q8: Orbital Distance in AU

Formula: $a = (P^2 M)^{1/3}$, with P in years. First convert: $P =$ days$/365.25$.

STAC-1 b ($M = 1.00\,M_\odot$)

Working:

$$P = 4/365.25 = 0.01095 \text{ yr}$$
$$a^3 = (0.01095)^2 \times 1.00 = 1.199 \times 10^{-4}$$
$$a = 0.0493 \text{ AU}$$

STAC-2 b ($M = 0.80\,M_\odot$)

Working:

$$P = 80/365.25 = 0.2190 \text{ yr}$$
$$a^3 = (0.2190)^2 \times 0.80 = 0.03838$$
$$a = 0.337 \text{ AU}$$

STAC-3 b ($M = 0.50\,M_\odot$)

Working:

$$P = 40/365.25 = 0.1095 \text{ yr}$$
$$a^3 = (0.1095)^2 \times 0.50 = 0.005997$$
$$a = 0.182 \text{ AU}$$

Answer: STAC-1 b: 0.049 AU. STAC-2 b: 0.337 AU. STAC-3 b: 0.182 AU.

B. Q9: Orbital Distance in Kilometres

Formula: AU $\times\ 149.6 \times 10^6$ km.

Working:

$$\text{STAC-1 b: } 0.0493 \times 149.6 \times 10^6 = 7.4 \times 10^6 \text{ km}$$
$$\text{STAC-2 b: } 0.337 \times 149.6 \times 10^6 = 5.05 \times 10^7 \text{ km}$$
$$\text{STAC-3 b: } 0.182 \times 149.6 \times 10^6 = 2.72 \times 10^7 \text{ km}$$

Answer: 7.4×10^6 km, 5.05×10^7 km and 2.72×10^7 km.

C. Q10: Comparison With Earth's Distance From the Sun

Earth is 1 AU from the Sun. We divide 1 AU by each orbital distance.

Working:

$$\text{STAC-1 b: } 1/0.0493 \approx 20 \text{ times closer}$$
$$\text{STAC-2 b: } 1/0.337 \approx 3.0 \text{ times closer}$$
$$\text{STAC-3 b: } 1/0.182 \approx 5.5 \text{ times closer}$$

Answer: All three planets orbit much closer to their stars than Earth does to the Sun. All three are also closer than Mercury is to the Sun (0.387 AU).

TABLE V
PARTS B AND C RESULTS

	STAC-1 b	STAC-2 b	STAC-3 b
Radius (km)	38,100	17,600	10,300
Radius ($R_\oplus$)	5.98	2.76	1.62
Class	Neptune-sized	Neptune-sized	Super-Earth
Orbital distance (AU)	0.049	0.337	0.182
Orbital distance (10^6 km)	7.4	50.5	27.2

V. PART D: FIND THE MOST PROMISING WORLD

A. Q11: Most Promising Candidate

First we find the habitable-zone distance of each star. *Formula:* $d_{\mathrm{HZ}} \approx \sqrt{L/L_\odot}$ AU.

We also find how much starlight each planet receives compared with Earth, using $S = (L/L_\odot)/a^2$.

Working:

$$d_{HZ,1} = \sqrt{1.00} = 1.000 \text{ AU}$$
$$d_{HZ,2} = \sqrt{0.40} = 0.632 \text{ AU}$$
$$d_{HZ,3} = \sqrt{0.08} = 0.283 \text{ AU}$$
$$S_1 = 1.00/(0.0493)^2 = 411$$
$$S_2 = 0.40/(0.337)^2 = 3.5$$
$$S_3 = 0.08/(0.182)^2 = 2.4$$

Then we compare the orbital distance a with d_{HZ}. A value close to 1 means the planet is close to the habitable zone.

TABLE VI
COMPARISON OF THE DETECTED PLANETS

	STAC-1 b	STAC-2 b	STAC-3 b
Orbital distance a (AU)	0.049	0.337	0.182
d_{HZ} (AU)	1.000	0.632	0.283
a/d_{HZ}	0.05	0.53	0.64
Starlight received ($\times$ Earth)	411	3.5	2.4
Radius ($R_\oplus$)	5.98	2.76	1.62

Answer: STAC-3 b is the most promising candidate for further habitability studies.

B. Q12: Evidence

1) **Closest to the habitable zone.** Its a/d_{HZ} is 0.64, the highest of the three. STAC-1 b has only 0.05 and receives about 411 times Earth's starlight, so it is far too hot.
2) **Smallest radius.** At $1.62\,R_\oplus$ it is a Super-Earth. Smaller planets are more likely to be rocky like Earth. The other two are Neptune-sized and probably made mostly of gas or ice.
3) **Moderate starlight.** It receives 2.4 times Earth's starlight, close to Venus (about 1.9 times). Clouds or an atmosphere could help keep it from getting too hot.
4) **Most transits observed.** Five transits were seen, so its orbital period and orbital distance are the most reliable.

Important: a planet being within the habitable zone does not necessarily mean that it is habitable. By our simple relation, STAC-3 b orbits slightly closer than d_{HZ}, so it may be a little hot. Its host star is small and red, so the planet may always keep the same side facing the star, and the star may have strong flares. Planets larger than about $1.6\,R_\oplus$ are often gassy, so it may not be rocky.

C. Q13: One Additional Observation

Answer: Transmission spectroscopy (for example with the James Webb Space Telescope). During a transit, a little starlight passes through the planet's atmosphere. The gases in the atmosphere absorb certain colours of light. By checking which colours are missing, we can find gases such as water vapour, carbon dioxide, methane or oxygen.

Measuring the mass of the planet would also help. With its radius, the mass gives the density, which shows whether the planet is rocky or gassy.

VI. CONCLUSION

All three stars host a transiting exoplanet. STAC-1 b ($P = 4$ days, $a = 0.049$ AU) is a large Neptune-sized planet. STAC-2 b ($P = 80$ days, $a = 0.337$ AU) is a smaller Neptune-sized planet. STAC-3 b ($P = 40$ days, $a = 0.182$ AU) is a Super-Earth and the most promising candidate for further study.

Limits: the observations are 1 to 20 days apart, so the orbital periods are approximate, especially for STAC-2. Real transits last only a few hours, so this data set is a simplified example.

VII. SIMPLE WORDS USED IN THIS REPORT

- **Exoplanet:** a planet that orbits a star other than the Sun.
- **Host star:** the star that a planet orbits.
- **Transit:** when a planet passes in front of its host star and blocks a little of its light.
- **Relative brightness:** the brightness of the star compared with its normal brightness (1.0000).
- **Transit depth (δ):** the fraction by which the star's brightness decreases during a transit.
- **Orbital period (P):** the time the planet takes to go around its star once.
- **Orbital distance (a):** how far the planet is from its star.
- **Luminosity (L):** the total light the star gives out.
- **Habitable zone:** the range of distances from a star where liquid water could exist on a planet.

REFERENCES

[1] J. N. Winn, "Transits and occultations," in *Exoplanets*, S. Seager, Ed. Tucson, AZ, USA: Univ. Arizona Press, 2010, pp. 55–77.

[2] NASA Exoplanet Archive. [Online]. Available: https://exoplanetarchive.ipac.caltech.edu